Royal Berkshire
NHS Foundation Trust

Obstetrics Antibiotic Guidelines (GL787)

Approval and Authorisation

Approval Group	Job Title, Chair of Committee	Date
Maternity & Children's Services Clinical Governance Committee	Chair Maternity Clinical Governance Committee	12th July 2024

Change History

Version	Date	Author(s)	Reason
Version 4.3	February 2019	Dr Shabnam Iyer Christiana Ogunmodede	Pg 17/18 - Inclusion of new guidance on Gentamicin monitoring
Version 4.4	June 2019	Dr Shabnam Iyer Christiana Ogunmodede	Changes to pg 13 to as agreed by Christine Harding, Mr Patrick Bose, Dr Shabnam Iyer and Christiana Ogunmodede following the results of Anode Study
Version 4.5	October 2019	Dr Shabnam Iyer Christiana Ogunmodede C Harding Dr L Williams	Live change to pg 12/13 regarding use of IV teicoplanin for penicillin allergic patients being treated for GBS and regimen to follow if patient proceeds to LSCS
Version 4.6	Dec 2019	Will Cooke (ST4 O&G)	Live change to include regimen for **Endometritis** following vaginal delivery (pg 10). Also additional changes made by C Ogunmodede in agreement with Dr Iyer
Version 5.0	May 2021	Dr Shabnam Iyer Wanqing Fan	Updated management of STIs. Additional contra-indications for nitrofurantoin. Updated antimicrobial prophylaxis recommendations for mid-term delivery in line with NICE guideline NG140: Abortion care (25.09.19) and, aligned the antimicrobial prophylaxis for GBS recommendations in preterm labour to NICE guideline NG195-Neonatal infection: antibiotics for prevention and treatment (21.04.21)
Version 5.1	June 2022	C Harding S Bisht Dr S Iyer	Pg 12/13 - Live change to ensure compliance with BAPM, NICE and MatNeo to give IV antibiotics to women in premature labour
Version 6.0	June 2024	Dr Shabnam Iyer Naveet Mahal	New gentamicin and teicoplanin dose banding tablets, (Appendices 1, 2 & 3), updated guidance for treatment of severe CAP in patients with anaphylactic-type penicillin allergy (Table 1 pg 8 & Table 2 pg 11), updated costs of antibiotics (Appendix 4) and new risk assessment for women at risk of pre-term birth (Appendix 6).

Contents

1. Consultation undertaken on this guideline

This guideline has been the subject of consultation with the Consultant Medical Microbiologist, Consultant Radiologist and Lead Antimicrobial Pharmacist. The guideline has also been considered and approved by the Maternity Clinical Governance Committee.

2. Implementation

This approved guideline will be placed on the intranet and will be cascaded though directors to line managers and staff.

3. Purpose of the Guideline

Is to provide evidence-based recommendations for the antibiotic treatment and prophylaxis of infections in obstetrics, based on the local epidemiology and antimicrobial susceptibility profiles of pathogens

4. Scope of the Guideline

It covers recommendations for:

1. Microbiological diagnosis and antibiotic treatment of common infections in pregnancy (during the antenatal, intrapartum and peripartum periods) and, postpartum period including the lactation period
2. Antibiotic prophylaxis for prevention of Early-onset Neonatal Group B Streptococcal disease (EOGBSD)
3. Antibiotic prophylaxis for common obstetric procedures

The guideline does not cover the antimicrobial management of genital herpes, chickenpox, parvovirus, CMV, hepatitis B/C or HIV. Please refer to the relevant obstetrics guidelines

The guideline should be used in conjunction with other relevant guidelines on the intranet such as:

- Adult Antimicrobial Guide on MicroGuide (mobile/web version)
- Management of penicillin allergy (GL991) on MicroGuide (mobile/web version)
- IV teicoplanin guideline (GL1122) on MicroGuide (mobile/web version)
- IV-to-Oral switch protocol (CG334) on MicroGuide (mobile/web version)
- Influenza guideline (GL1145) on MicroGuide (mobile/web version)
- General Principles of Surgical Antimicrobial Prophylaxis on MicroGuide (mobile/web version)
- MRSA Screening protocol (CG179) on Policy hub
- Maternal sepsis prevention, recognition and management guideline (GL872) on Policy hub
- Group B Streptococcal Infection (GBS) guideline (GL850) on Policy Hub

5. Acronyms and abbreviations

APN: Acute pyelonephritis
ASB: Asymptomatic bacteriuria
ARDS: Acute Respiratory Distress Syndrome
BC: Blood culture
CCU: Clean catch urine
CFU: Colony forming unit
CMM: Consultant Medical Microbiologist
CT: *Chlamydia trachomatis*
CT/NG PCR: *Chlamydia trachomatis* and *Neisseria gonorrhoeae* PCR
Cx: Cervical swab
eGFR: Estimated glomerular filtration rate
EOGBSD: Early-onset Group B Streptococcus disease
ESBL: Extended-spectrum beta-lactamase
GAS: Group A Streptococcus (*Streptococcus pyogenes*)
GBS: Group B streptococcus (*Streptococcus agalactiae*)
GC: Gonococcus (*Neisseria gonorrhoeae*)
GNB: Gram-negative bacteria
GUM: Genitourinary medicine
G-6PD: Glucose-6-Phosphate Dehydrogenase
HIV: Human immunodeficiency virus
HVS: High vaginal swab
IAI: Intra-amniotic infection (chorioamnionitis)
IAP: Intrapartum antibiotic prophylaxis
I&D: Incision and drainage
LBW: Low birth weight
LE: Leucocyte esterase
LN: Lymph node
BP: Blood pressure
CDI: *Clostridium difficile* infection
C&S: Culture and sensitivity
CA-MRSA: Community-associated MRSA
LSCS: Lower Segment Caesarean section
LVS: Low vaginal swab
MRSA: Methicillin-resistant *Staphylococcus aureus*
MSU: Midstream urine
N: Nitrites
NG: *Neisseria gonorrhoeae*
PCR: Polymerase chain reaction
PL: Pre-labour
PLROM: Pre-labour rupture of membranes
POC: Product of conception
PR: Pulse rate
PROM: Premature rupture of membranes
P-PLROM: Pre-term prelabour rupture of membranes
P-PROM: Pre-term premature rupture of membranes
O/e: On examination
RBCs: Red blood cells
RR: Respiratory rate
SOB: Shortness of breath
SpO_2: Oxygen saturations
STI: Sexually Transmitted Infection
ST-LVS: Self-taken low vaginal swab
TDM: Therapeutic drug monitoring
TSS: Toxic shock syndrome
UC&S: Urine culture and sensitivity
UTI: Urinary tract infection
VTM: Viral transport medium

6. Terms and definitions

a) **Bacteriuria** defined as presence of bacteria in urine, can be detected by:
- Urine dipstick test for nitrites (N), resulting from conversion of dietary nitrates to nitrites by 'coliform' bacteria (*Enterobacteriaceae*).
    - At least 4 hour bladder incubation is required for N test to be reliable.
    - Nitrite reagent on the dipstick is sensitive to environmental air- improperly stored, or use of out-of-date dipsticks could result in false-positive N test.
    - Non-nitrate reducing organism and low nitrate diet could produce a false-negative N result.
- Urine culture is the gold standard (reference) test for detection of bacteriuria
- **Asymptomatic bacteriuria (ASB)** refers to presence of $\geq 10^5$ CFU/mL of bacteria in MSU in absence of symptoms of UTI

b) **CCU** refers to specimen of urine collected after local cleaning of urethral meatus and surrounding mucosa.

c) **Early-onset neonatal group B streptococcus disease (EOGBSD)** is defined as GBS infection with onset within 72 hours of birth

d) **ESBL-**producing GNB refers to a strain of GNBs that produces beta-lactamase enzymes capable of hydrolysing broad-spectrum cephalosporin's such as Cefotaxime, ceftriaxone and Ceftazidime

e) **Haematuria** defined as presence of >2 RBCs/mm^3, can be detected by urine dipstick test for haematuria (H)

f) **'High risk for GBS'** refers to a woman with:

- Previous baby with early-or late-onset invasive GBS infection.
- GBS colonisation detected incidentally or by intentional testing of vaginal or perineal swab (specimen)
- GBS bacteriuria or infection in the current pregnancy

g) **'High risk for MRSA'** refers to a patient with ≥ 1 of following risk factors
1. Known 'MRSA Positive' (previously infected or colonised with MRSA)
2. Hospitalisation in or outside the trust in the past year
3. Frequent re-admissions to any healthcare facilities
4. Diabetes and BMI ≥ 40 kg/m^2
5. **Health care worker** with any form of direct patient contact and veterinary staff
6. Intravenous drug use
7. Patients infected with HIV

h) **'High risk for STIs'**
8. ≤25 years old
9. New sexual partner or >1 sexual partner in the last 12 months, or sex partner with concurrent partners
10. Sex contact of a case with STI
11. Asymptomatic women requesting 'STI Screen'

i) **Hospital-acquired infection (HAI)** refers to an infection that is neither present nor incubating at the time of hospital admission but, acquired ≥ 48 hours after admission

j) **Intra-amniotic infection (IAI) or chorioamnionitis** refers to infection of foetal membranes (chorion and amnion) amniotic fluid, foetus, umbilical cord and placenta**.**

k) **MSU** refers to second portion of the voided urine specimen collected after discarding the initial stream.

l) **Modified Early Obstetric Warning Score (MEOWS)** refers to screening tool used for maternal sepsis. Please refer to the **Maternal sepsis prevention, recognition and management (GL872)**

m) **Multidrug resistant (MDR) bacteria** refers to strains of bacteria that are resistant to ≥2 classes of antibiotics such as MRSA is resistant to all beta lactam class of antibiotics and may also be resistant to macrolide antibiotics such as erythromycin and quinolone class of antibiotics such as ciprofloxacin

n) **Preterm** refers to before 37 completed weeks of gestation

o) **Preterm labour** refers to onset of labour before 37 completed weeks of gestation

p) **Premature rupture of membranes (PROM)** refers to rupture of membranes before onset of labour or prelabour ROM **(PLROM)**

q) **Preterm PROM (PPROM) or PPLROM** refers to PROM before 37 completed weeks of gestation

r) **Pyuria** refers to presence of white cells (pus cells) in urine specimen, can be detected by:
   - Urine dipstick test for leucocyte esterase **(LE)**
     Urine dipstick test for LE is more sensitive than, and as specific as urine microscopy for detection of pyuria.

s) **Sepsis** is defined as the life-threatening organ dysfunction resulting from dysregulated host response to infection

t) **Surgical Antimicrobial Prophylaxis (SAMP)** is defined as the peri-procedural administration of a **<u>single therapeutic IV</u>** dose of an antimicrobial agent, administered usually **<u>30-60 minutes before the start of a procedure</u>** to prevent infectious complication

Table 1: Empirical Antibiotic Treatment Recommendations

BMI-based patient stratification: Lean (BMI <30), Obese (BMI 30-39.9), morbidly obese (BMI≥40). Use patient's antenatal booking weight to calculate BMI

All dosage recommendations are for patients with normal renal and hepatic functions unless stated otherwise. Dosages should be adjusted to suit person's age, weight, hepatic and renal function. Please contact your ward pharmacist, if necessary.

<table>
<tr><th rowspan="2">Clinical Indication</th><th rowspan="2">Clinical Assessment</th><th rowspan="2">Diagnostic Evaluation</th><th colspan="2">Treatment Recommendations</th><th rowspan="2">Comments</th></tr>
<tr><th>First Choice</th><th>Alternatives e.g. 'True Penicillin allergy' (See penicillin allergy guideline here)</th></tr>
<tr><td rowspan="5">Asymptomatic bacteriuria (ASB)</td>
<td rowspan="5">• All intermediate and high risk pregnant women should be screened for ASB (with urine culture) at 12-16 weeks gestation or at the first antenatal visit (Appendix 6)

• ASB is diagnosed by positive urine culture (with bacterial growth $\geq 10^8$ CFU/L) <u>WITHOUT</u> any symptoms of UTI</td>
<td rowspan="5">• Urine dipstick tests for LE and N should not be used

• MSU should be sent for C&S

• A follow-up urine culture as test of cure (TOC) should be done a week after completion of antimicrobial treatment:
▪ <u>Persistent ASB</u>: If the TOC urine culture is positive with the same organism
▪ <u>Recurrent ASB</u>: If the TOC urine culture is positive with a different organism or TOC urine culture is negative but a subsequent urine culture is positive with the same or different organism</td>
<td colspan="2"><u>Always treat based on the UC&S results</u></td>
<td rowspan="5">• ASB in pregnancy <u>MUST</u> be treated to reduce risk of progression to acute pyelonephritis, low birth weight, pre-term delivery and perinatal mortality</td></tr>
<tr><td>• PO amoxicillin 1g 8 hourly <u>for 3 days</u>
<u>OR</u>
• PO cefradine 500mg 6 hourly <u>for 3 days</u></td>
<td>• PO nitrofurantoin 50mg 6 hourly <u>for 3 days</u> (if eGFR ≥ 45mL/min)
(Avoid in the third trimester and at term, in G-6PD deficiency and in acute porphyrias), <u>OR</u>
• PO trimethoprim 200mg 12 hourly <u>for 3 days</u>
(Avoid in the first trimester
Avoid in those on a folate antagonist
And consider 5mg folic acid daily in those at risk of folic acid deficiency)</td></tr>
<tr><td colspan="2"><u>MDR infection</u>: Based on the UC&S results
• PO fosfomycin 3g <u>single dose</u>, <u>OR</u>
• PO nitrofurantoin 50mg 6 hourly <u>for 3 days</u> (if eGFR ≥ 45mL/min)
Avoid use in the third trimester and at term, in G-6PD deficiency and in acute porphyrias)
• Please discuss with CMM, if required</td></tr>
<tr><td colspan="2"><u>Persistent ASB</u>:
• Repeat another course of antimicrobial treatment (as above) based on UC&S</td></tr>
<tr><td colspan="2"><u>Recurrent ASB:</u>
• Treat based on UC&S
• Antimicrobial prophylaxis is <u>NOT</u> recommended</td></tr>
<tr><td>Cystitis</td>
<td>• Dysuria, frequency/urgency, supra-pubic discomfort
• Fever and chills are generally absent in isolated cystitis</td>
<td>• Urine dipstick for LE and N
• MSU/CCU for C &S
• Diagnosis of cystitis is confirmed by positive urine culture with bacterial growth of $\geq 10^4$-10^5 CFU/mL</td>
<td colspan="2">• Please review previous UC&S results. Discuss with CMM, if necessary
• Treat as above for ASB

<u>Recurrent UTI</u>:
• PO cefradine 500mg (post-coital or at bedtime)
<u>OR</u>
• PO nitrofurantoin 50mg if eGFR ≥ 45mL/min (post-coital or at bedtime)
(Avoid use during the third trimester and at term, in G-6PD deficiency and in acute porphyrias)</td>
<td>• Dysuria without positive culture (bacteriuria) or persistent dysuria despite successful antimicrobial treatment of bacteriuria should warrant testing for STIs (CT & NG)</td></tr>
</table>

Clinical Indication	Clinical Assessment	Diagnostic Evaluation	Treatment Recommendations: First Choice	Treatment Recommendations: Alternatives e.g. 'True Penicillin allergy' (See penicillin allergy guideline here)	Comments
Acute Pyelonephritis (APN)	• Fever and chills, nausea/vomiting, flank pain renal angle tenderness • ± symptoms or signs (s/s) of cystitis	• Urine dipstick for LE and N • MSU/CCU for C &S • BC (before antibiotics)	IV aztreonam 2g 8 hourly **Known ESBL Positive**: • IV temocillin 2g 12 hourly for 24-48 hours followed by IV/PO treatment based on the UC&S • **Duration of treatment: 7 days**	• IV aztreonam 2g 8 hourly	• Once afebrile for 48 hours switch IV to PO antibiotics based on UC&S
Urosepsis	• UTI + MEOWS ≥3	• BC (before antibiotics) • MSU/CCU for C &S	• IV Meropenem 1g 8 hourly • Review and revise treatment in light of culture results	• IV gentamicin* 5mg/kg OD	***See Appendix 1&2 for information on gentamicin monitoring and dosing***
Pharyngitis	• Fever, tonsillar exudate, tender anterior cervical LNs	• Throat swab for C&S	• PO amoxicillin 1g 8 hourly **OR** IV amoxicillin 1g 6 hourly	• PO clarithromycin 500mg 12 hourly (Avoid in 1st trimester unless benefit outweighs risk)	
Community acquired pneumonia (CAP)	• Fever, SOB, cough +/-sputum, chest pain • Antecedent history of (h/o) influenza	• **BC** (before antibiotics) • Sputum for C&S • Pneumococcal & Legionella urine antigen tests • **In Winter months:** Nasopharyngeal swab in VTM for Flu PCR	• PO/IV amoxicillin 1g 8 hourly + • PO/IV azithromycin 500mg 24 hourly	**Non-anaphylactic type-Pen-allergy** • IV ceftriaxone 2g 24 hourly + • PO/IV azithromycin 500mg 24 hourly **Anaphylactic type Pen-allergy** • IV teicoplanin** 10mg/kg 12 hourly for 3 doses, then 10mg/kg 24 hourly + • PO/IV azithromycin 500mg 24 hourly	• Pregnant women with pneumonia are at risk of preterm labour/delivery • In-patient treatment until clinical stability is recommended • Switch IV ceftriaxone 2g 24 hourly to PO cefradine 500mg 6 hourly when clinical improvement ***See Appendix 3 for information on teicoplanin dosing.***
	Severe CAP: • Rapidly progressive CAP evolving into ARDS • Severe sepsis • Haemoptysis		**Severe CAP** • IV benzylpenicillin 1.2g 6 hourly + • IV clindamycin 1.2g 6 hourly OR PO clindamycin 450mg 6 hourly	**Non-anaphylactic type-Pen-allergy** • IV ceftriaxone 2g 24 hourly + • IV clindamycin 1.2g 6 hourly OR PO clindamycin 450mg 6 hourly **Anaphylactic type Pen-allergy** • IV teicoplanin** 10mg/kg 12 hourly for 3 doses, then 10mg/kg 24 hourly + • IV clindamycin 1.2g 6 hourly OR PO clindamycin 450mg 6 hourly	

<table>
<tr><th rowspan="2">Clinical Indication</th><th rowspan="2">Clinical Assessment</th><th rowspan="2">Diagnostic Evaluation</th><th colspan="2">Treatment Recommendations</th><th rowspan="2">Comments</th></tr>
<tr><th>First Choice</th><th>Alternatives e.g. 'True Penicillin allergy' (See penicillin allergy guideline here)</th></tr>
<tr><td>Clostridium difficile Infection</td><td></td><td>• Diarrhoeal stool specimen for CDI testing</td><td colspan="3">Please refer to the Clostridioides (formerly Clostridium) difficile infection on Micro Guide (mobile/web version)</td></tr>
<tr><td rowspan="2">Maternal Sepsis:
▪ Sepsis in pregnancy
▪ Chorioamnionitis or IAI
▪ Endometritis
▪ Peri-partum maternal pyrexia
▪ Sepsis following pregnancy (≤6 weeks postnatal)
▪ Post-partum sepsis/post-abortion sepsis/puerperal sepsis
▪ Post-partum endometritis
▪ Sepsis of unknown origin</td><td rowspan="2">S/s relevant to the most likely focus of infection such as:
• Pharyngitis
• Pneumonia
• UTI
• IAI
• Breast abscess: Fever, rigors, spiking temperature
• Mastitis: Breast engorgement/redness
• TSS: Generalised maculopapular rash
• Endometritis: Abdominal pain/tenderness, offensive vaginal discharge
• Pneumonia: Cough, SOB, chest pain
• Wound infection: LSCS, episiotomy, perineal wound, epidural site
• Gastrointestinal (GI): Diarrhoea, vomiting
• Non-specific s/s: lethargy, reduced appetite
<u>PLUS</u>
• MEOWS ≥3:
▪ Temperature, PR, RR, BP, level of consciousness, SpO2
• Risk stratify sepsis as:
▪ Red flag or Amber flag sepsis
▪ Or, low risk of sepsis</td><td rowspan="2">• BC X 2 sets (prior to antibiotics)
• Other relevant cultures based on the suspected focus of infection e.g:
▪ IAI: Amniotic fluid, HVS
▪ Endometritis: HVS
▪ Post-partum endometritis: LVS placental swab
▪ Meningitis: Cerebrospinal fluid (CSF)
▪ Epidural abscess
▪ Mastitis: expressed breast milk
▪ Caesarean section or perineal wound swab
▪ Faeces, if diarrhoea
▪ UC&S, if UTI</td><td>First Choice
• IV amoxicillin 1g 8 hourly
+
IV clindamycin 1.2g 6 hourly for 48 – 72 hours
+
IV gentamicin* 5mg/kg STAT

Followed by:
• Pathogen-targeted therapy based on the C&S results and clinical re-assessment
• Discuss with CMM, if necessary</td><td><u>MRSA Positive or Penicillin allergy:</u>
• IV teicoplanin** 10mg/kg 12 hourly for <u>3 doses</u>, then 10mg/kg 24 hourly
+
IV clindamycin 1.2g 6 hourly
+
IV gentamicin* 5mg/kg STAT

Followed by:
• Pathogen-targeted therapy based on the C&S results and clinical re-assessment
• Discuss with CMM, if necessary</td><td rowspan="2">See Appendix 1&2 for information on gentamicin monitoring and dosing.

See Appendix 3 for information on teicoplanin dosing.</td></tr>
<tr><td>Known ESBL positive
• IV temocillin 2g 12 hourly
+
• IV clindamycin 1.2g 6 hourly

For 48 – 72 hours, followed by:
• Pathogen-targeted therapy based on the C&S results and clinical re-assessment
• Discuss with CMM, if necessary</td><td>Known ESBL positive
• IV teicoplanin** 10mg/kg 12 hourly for <u>3 doses</u>, then 10mg/kg 24 hourly
+
• IV clindamycin 1.2g 6 hourly
+
• IV gentamicin* 5mg/kg <u>STAT</u>

Followed by:
• Pathogen-targeted therapy based on the C&S results and clinical re-assessment
• Discuss with CMM, if necessary</td></tr>
</table>

Clinical Indication	Clinical Assessment	Diagnostic Evaluation	Treatment Recommendations		Comments
			First Choice	Alternatives e.g. 'True Penicillin allergy' (See penicillin allergy guideline here)	
Caesarean section (incisional) wound infection	• S/s of superficial incisional wound infection: redness, pain, swelling, wound dehiscence, discharge • S/s deep tissue infection: haematomas, seroma, abscess • S/s of post-caesarean section endometritis	• Wound swab or aspirate	• Antibiotic treatment is recommended for localised cellulitis or associated sepsis or associated sepsis in addition to drainage of collection or debridement • Surgical review of wound to drain any localised or deep collection		• Caesarean section wound infection is a surgical site infection. ***See Appendix 3 for information on teicoplanin dosing.***
			• IV co-amoxiclav 1.2g 8 hourly <u>OR</u> PO co-amoxiclav 625mg 8 hourly	<u>**Penicillin-allergy or MRSA Positive**</u> • IV teicoplanin** 10mg/kg 12 hourly <u>for 3 doses</u>, then 10mg/kg 24 hourly	
Mild – Moderate Endometritis following vaginal delivery. **NB: If severe treat as MATERNAL SEPSIS**	• S+S of endometritis post vaginal delivery	• Vaginal swab	• PO co-amoxiclav 62f5mg 8 hourly	• IV gentamicin* 5mg/kg for 48hrs **+** • PO clindamycin 450mg 6 hourly <u>**for 5 days**</u>	Re-evaluate penicillin allergy status. Only consider alternative treatment if patient has true penicillin allergy. ***See Appendix 1&2 for information on gentamicin monitoring and dosing.***
▪ **Lactational mastitis** ▪ **Breast abscess**	• Red, swollen, painful breast **±** • S/s of breast abscess: Localised painful fluctuant and tender mass associated with fever, rigors and malaise	• Expressed milk for C&S if severe sepsis, hospital-acquired infection (HAI) or unresponsive to initial treatment • **BC** X 2 sets if clinically 'septic' • Pus/aspirate for C&S	PO flucloxacillin 1g 6 hourly	IV clindamycin 1.2g 6 hourly <u>OR</u> PO clindamycin 450mg 6 hourly	***See Appendix 3 for information on teicoplanin dosing.***
			<u>**Known MRSA positive**</u> • IV teicoplanin** 10mg/kg 12 hourly for <u>3 doses</u>, then 10mg/kg 24 hourly <u>**OR**</u> • PO linezolid 600mg 12 hourly <u>**for 5-7 days**</u> (*Use only if no other option and benefit outweighs risk*)		

***Use booking weight for BMI < 30. Use corrected dosing weight if BMI ≥ 30. See appendix 1&2 below to calculate gentamicin dose**

****Use booking weight for actual weight**

Table 2: Out-patient Management of Common Infections in Pregnancy

All dosage recommendations are for patients with normal renal and hepatic functions unless stated otherwise. Dosages should be adjusted to suit person's age, weight, hepatic and renal function. *Please contact your pharmacist, if necessary.*

Clinical condition/diagnosis	Microbiology Investigation	First-line treatment	Alternative(s)
Urinary tract infection (UTI) *For empirical treatment please review previous UC&S results*, if any	**Always send** • MSU/CCU for C &S	PO nitrofurantoin 50mg 6 hourly **for 7 days** (if eGFR ≥ 45mL/min) *(Avoid in the third trimester and at term, in G-6PD deficiency and in acute porphyrias)* **OR if eGFR <45mL/min AND** **If sensitive to** amoxicillin/cefradine: • PO amoxicillin 1g 8 hourly **for 7 days** • PO cefradine 500mg 6 hourly **for 7 days**	**If sensitive to** trimethoprim • PO trimethoprim 200mg 12 hourly **for 7 days** *(Avoid in the first trimester* *Avoid in those on a folate antagonist* *And consider 5mg folic acid daily in those at risk of folic acid deficiency)*
Community acquired pneumonia (CAP) **'Low risk' (CRB65= 0)**		• PO amoxicillin 1g 8 hourly **for 5 days**	**Non-anaphylactic type-Pen-allergy** • PO cefradine 500mg 6 hourly **for 5 days** **Anaphylactic type Pen-allergy** • PO clindamycin 450mg 6 hourly
Abnormal vaginal discharge: Common causes of abnormal vaginal discharge include: Bacterial vaginosis (BV), Vulvovaginal candidiasis (VVC), Trichomonas vaginalis (TV) and other sexually transmitted infection (STIs). Please refer to PHE Primary care guidance on management and laboratory diagnosis of Abnormal Vaginal Discharge **HVS culture is indicated in cases of:** • Recurrent symptoms/treatment failure • Inconclusive assessment • Pre-or post-termination of pregnancy • Postnatal or post-miscarriage			
• **Bacterial vaginosis (BV)** *BV is diagnosed clinically based on the following symptoms and signs (s/s):* • *Offensive fishy smelling vaginal discharge* • *Not associated with soreness, itching, or irritation* • *Thin, white, homogenous discharge, coating the walls of the vagina or vestibule*	• Routine HVS culture is not recommended for laboratory diagnosis of BV. • You may choose to use a narrow-range pH paper (pH 3.8-5.5) to test for the pH of the vaginal discharge at the time of examination for diagnosis of BV and to distinguish between BV and other causes of discharge. • Urine PH dipsticks are not suitable. • If symptoms are typical, there is no itch or soreness, and vaginal pH > 4.5, there is no need for a swab for microscopy and/or culture	PO metronidazole 400mg 12 hourly **for 5 days**	**Breast feeding** • Intravaginal metronidazole gel (0.75%) once daily at night **for 5 days** **OR** • Intravaginal clindamycin cream (2%) once daily at night **for 7 days**

Clinical condition/diagnosis	Microbiology Investigation	First-line treatment	Alternative(s)
• **Vulvovaginal candidiasis (VVC) ('Thrush')**	• Culture not required unless recurrent	Intra-vaginal clotrimazole 500mg STAT + topical clotrimazole 1% cream 12 hourly **for 10 days**	
• **Suspected (STIs)**: Gonorrhoea (NG)/Chlamydia (CT)/Trichomonas vaginalis (TV)	• Self-taken vaginal swab or low/high vaginal swab in Chlamydia transport medium for CT/NG/TV PCR	**Please refer to GUM in ALL cases** IM ceftriaxone 1g **single dose** **+** PO azithromycin 1g **single dose** **+** PO metronidazole 400mg 12 hourly **for 5 days**	
Mastitis (*If breast* abscess: Please refer patient to hospital for I&D/aspiration of abscess)	• Pus/aspirate	PO flucloxacillin 1g 6 hourly **for 5 days**	PO clindamycin 450mg 6 hourly **for 5 days**

Table 3: Antimicrobial Prophylaxis Recommendations

BMI-based patient stratification: Lean (BMI <30), Obese (BMI 30-39.9), morbidly obese (BMI≥40). Use patient's antenatal booking weight to calculate BMI

All dosage recommendations are for patients with normal renal and hepatic functions unless stated otherwise. Dosages should be adjusted to suit person's age, weight, hepatic and renal function. *Please contact your ward pharmacist, if necessary*.

Clinical Indication	Rationale	First Choice	Alternative: Penicillin allergy *(Please refer to 'management of penicillin allergy' protocol)*
Amniocentesis	• **Antibiotic prophylaxis is NOT recommended**		

Evacuation of retained products of conception (ERPC) after medical termination of pregnancy or mid-trimester delivery	• To reduce risk of upper genital tract infection	• All women should be screened for STI (CT/NG)
		  • **CT-Positive**: PO doxycycline 100mg 12 hourly for **7 days**
		  • **CT Negative**: metronidazole 1g rectally **single dose**
		  • **Not screened/STI screening result unavailable**: PO doxycycline 100mg 12 hourly for **3 days**	

Surgical management of miscarriage in first trimester	• To reduce risk of upper genital tract infection	Metronidazole 1g rectally **single dose** + PO doxycycline 100mg 12 hourly **for 3 days**	
Pre-term premature rupture of membrane (P-PROM) or Preterm pre-labour rupture of membranes (P-PLROM)	• To prolong pregnancy • To reduce maternal and neonatal infection and, • Gestational age-related neonatal morbidity	• PO amoxicillin 1g 8 hourly **for 7 days** or until the women is in established labour (whichever is sooner) • In labour require antibiotics as for GBS (see below)	• PO azithromycin 500mg 24 hourly **for 7 days** or until the women is in established labour (whichever is sooner)
Cervical cerclage	• To reduce risk of endometritis	• IV co-amoxiclav 1.2g **single dose**	• IV clindamycin 1.2g **single dose**

Premature/prelabour rupture of membranes (PROM/PLROM) at term	• To prevent Early-onset Neonatal Group B Streptococcal disease (EOGBSD)	• **Routine antibiotic prophylaxis is not recommended for women with premature/prelabour rupture of amniotic membranes at term**
		• **Antibiotic prophylaxis** (as above for P-PROM or P-PLROM) **is indicated if:**
		  • 'GBS Positive' i.e., GBS colonisation, bacteriuria or infection detected in the current pregnancy
		  • Prolonged rupture of membranes ≥ 48 hours
		  • Maternal pyrexia (Temperature >38°C)	

Preterm labour **Antibiotic prophylaxis is recommended** **If evidence of infection** e.g., maternal pyrexia (Temperature >38^{o}C): **Treat *as above* for IAI or chorioamnionitis**	• IV benzylpenicillin 3g **single dose** then 1.5g 4 hourly until delivered		**For penicillin allergic patients use:** • **IV teicoplanin* 10mg/kg STAT dose and then 10mg/kg every 12 hours until delivered** ***(See Appendix 3 for information on teicoplanin dosing)*** • **If patient proceeds to have LSCS, *add gentamicin** 5mg/kg STAT*** ***(See Appendix 1&2 for information on gentamicin monitoring and dosing)***
Examination under anaesthetic (EUA)	• To reduce risk of endometritis		

Clinical Indication	Rationale	First Choice	Alternative: Penicillin allergy *(Please refer to 'management of penicillin allergy' protocol)*
<u>IAP for GBS is indicated for women with:</u> • Previous baby with early-or late-onset invasive GBS infection. • GBS colonisation detected incidentally or by intentional testing of vaginal or perineal swab (specimen), GBS bacteriuria or infection, **<u>in the current pregnancy</u>**: ▪ **<u>Universal antenatal screening for GBS is not recommended</u>** ▪ Women with GBS detected in a previous pregnancy should be offered the choice of IAP without testing or, a bacteriological screening for GBS at 35-37 weeks or 3-5 days before the anticipated delivery date, and IAP if GBS carriage is detected in the current pregnancy **<u>Additional IAP for GBS is not required for women:</u>** • Undergoing Caesarean delivery (TREAT AS PER CAESAREAN SECTION GUIDANCE BELOW) • Receiving treatment with broad spectrum antibiotics for pyrexia in labour or chorioamnionitis	• To prevent Early-onset Neonatal Group B Streptococcal disease (EOGBSD)	• IV benzylpenicillin 3g **<u>single dose</u>** then 1.5g 4 hourly until delivered	**For penicillin allergic patients use:** • **IV teicoplanin* 10mg/kg STAT dose and then 10mg/kg every 12 hours until delivered** ***(See <u>Appendix 3</u> for information on teicoplanin dosing)*** • **If patient proceeds to have LSCS, *<u>add gentamicin** 5mg/kg STAT</u>*** ***(See <u>Appendix 1&2</u> for information on gentamicin monitoring and dosing)***
Episiotomy repair following normal vaginal delivery	• **Routine antibiotic prophylaxis before or immediately after incision or repair of episiotomy for women with uncomplicated vaginal delivery is <u>NOT RECOMMENDED</u>**		
Operative vaginal delivery (with either forceps or vacuum-assisted delivery)	**First Choice- No h/o Penicillin allergy**		**Alternatives e.g. 'True Penicillin allergy'**
	• **BMI < 30**: IV co-amoxiclav 1.2g **<u>single dose</u>** • **BMI ≥ 30:** (IV co-amoxiclav 1.2g + IV amoxicillin 1g) **<u>single dose</u>** **No further doses of antibiotics required**		• IV clindamycin 1.2g **<u>single dose</u>** + IV gentamicin** 5mg/kg **<u>STAT</u>** *(See <u>Appendix 1&2</u> for information on gentamicin monitoring and dosing)* **No further doses of antibiotics required**
Repair of third-and fourth degree perineal injuries	• To reduce risk of perineal wound infection, wound dehiscence or wound discharge	• **Routine antibiotic prophylaxis is not recommended for women with a third- or fourth-degree perineal tear**	
		• **BMI < 30**: IV co-amoxiclav 1.2g **<u>single dose</u>** • **BMI ≥ 30:** (IV co-amoxiclav 1.2g + IV amoxicillin 1g) **<u>single dose</u>** **No further doses of antibiotics required**	• IV clindamycin 1.2g **<u>single dose</u>** + IV gentamicin** 5mg/kg **<u>STAT</u>** *(See <u>Appendix 1&2</u> for information on gentamicin monitoring and dosing)* **No further doses of antibiotics required**

Clinical Indication	Rationale	First Choice	Alternative: Penicillin allergy *(Please refer to 'management of penicillin allergy' protocol)*

Caesarean section
No risk for MRSA
	• To reduce risk of surgical site infection	• **Routine antibiotic prophylaxis is recommended for women undergoing elective or emergency caesarean section**
		  • Prophylactic antibiotics should be administered peri-operatively, usually 30 minutes before skin incision
		  • **In emergency LSCS, in patients with PPROM or PROM over 4h or cervical dilatation ≥4 cm-Please use additional prophylaxis with IV azithromycin 500mg Single-dose (see flowchart on pg 24)**	

		• **BMI < 30**: IV co-amoxiclav 1.2g **single dose** • **BMI ≥ 30**: (IV co-amoxiclav 1.2g + IV amoxicillin 1g) **single dose**	• IV clindamycin 1.2g **STAT** + IV gentamicin** 5mg/kg **STAT** ***(See Appendix 1&2 for information on gentamicin monitoring and dosing)***
'Known MRSA' or 'High risk for MRSA'		• **IV** teicoplanin* 10mg/kg **STAT** + IV gentamicin** 5mg/kg **STAT** ***(See Appendix 1&2 for information on gentamicin monitoring and dosing)*** ***(See Appendix 3 for information on teicoplanin dosing)***	

Manual removal of placenta	• To reduce risk of postpartum endometritis	• **Routine antibiotic prophylaxis is recommended for women undergoing manual removal of the placenta**
		  • At Caesarean delivery: No additional antimicrobial prophylaxis is necessary
		  • After vaginal delivery: *As above* for Caesarean section (no risk for MRSA)	

Postpartum dilatation and curettage for retained products of conception	• **Antibiotic prophylaxis is NOT recommended**		

*** Use booking weight for actual weight**

**** Use booking weight for BMI < 30. Use corrected dosing weight if BMI ≥ 30. See appendix 1&2 below to calculate gentamicin dose**

If BMI ≥ 30: Corrected dosing weight = Ideal Body Weight + 40% of excess body weight (booking weight – ideal body weight)

Click for link to adult IV gentamicin guide

Click for link to adult IV teicoplanin guide

Table 4: Specific condition/pathogen-directed treatment recommendations

BMI-based patient stratification: Lean (BMI <30), Obese (BMI 30-39.9), morbidly obese (BMI≥40). Use patient's antenatal booking weight to calculate BMI

All dosage recommendations are for patients with normal renal and hepatic functions unless stated otherwise. Dosages should be adjusted to suit person's age, weight, hepatic and renal function. *Please contact your ward pharmacist, if necessary.*

Clinical Indication	Clinical Assessment	Diagnostic Evaluation	Treatment Recommendations: First Choice	Treatment Recommendations: Alternatives e.g. 'True Penicillin allergy'	Comments
Vaginal discharge in pregnancy/post-partum		• HVS ± Endocervical swab C&S **'High risk for STI's** or clinical suspicion of STI's : **CT/NG PCR** **Symptomatic patient** • HVS ± Endocervical swab • CCU **Asymptomatic patient** • CCU	**GAS**: • PO amoxicillin 1g 8 hourly **for 5 days** **OR** • PO cefradine 500mg 6 hourly **for 5 days** **OR** • IV benzylpenicillin 1.2g 4 hourly + IV clindamycin 1.2g 6 hourly	• PO clindamycin 450mg 6 hourly **for 5 days** **OR** • IV clindamycin 1.2g 6 hourly	• IAP (as above) is indicated if GBS is detected on a vaginal swab in the current pregnancy
Trichomonas vaginalis (TV)	• Vaginitis associated with frothy, greenish-yellow discharge, vulval soreness/itching, dysuria • TV is almost exclusively an STI. • TV is associated with high prevalence of co-infection with other STIs. Please refer to GUM	• HVS for smear microscopy • HVS/Cx/CCU for CT/NG/HSV & TV PCR	• PO metronidazole 400mg 12 hourly **for 5 days** (Metronidazole is safe to use in pregnancy) **Breast feeding** • Breastfeeding should be withheld during treatment and for 12-24 hours after the last dose to reduce infant exposure to metronidazole	• There is no suitable alternative to PO metronidazole to treat TV • Safety of tinidazole in pregnancy has not been evaluated	• Screening of sexual contacts for all STIs and treatment for TV irrespective of results is indicated • Routine screening or treatment of asymptomatic women for TV to reduce perinatal complications is not indicated • Asymptomatic women may be treated after 37 weeks to prevent perinatal transmission and reduce risk of HIV acquisition and transmission
Vaginal Candidiasis	Non-offensive white curdy discharge Vulval itch/soreness/Dysuria O/e: Vulval erythema, fissuring, oedema, excoriation	• HVS, LVS	Clotrimazole 500mg PV **STAT** + topical clotrimazole 1% cream 12 hourly **for 10 days**		Culture not required unless recurrent

Clinical Indication	Clinical Assessment	Diagnostic Evaluation	Treatment Recommendations: First Choice	Treatment Recommendations: Alternatives e.g. 'True Penicillin allergy'	Comments
Bacterial Vaginosis (BV)	• Thin, grey/white, homogenous discharge coating the walls of the vagina and vestibule • Fishy/offensive odour • Not associated with soreness, itching or inflammation	• Vaginal pH > 4.5 using narrow range pH paper (pH 3.8-5.5) **Microscopy:** • Wet mount microscopy for Clue cells performed in GUM clinic, or • A smear of discharge sent to lab for gram-stain Microscopy indicated if: ○ Recurrent or, ○ 'High risk for STIs'	• PO metronidazole 400mg 12 hourly **for 5** days	**Breast feeding** • Intravaginal metronidazole gel (0.75%) once daily at night **for 5 days** **OR** • Intravaginal clindamycin cream (2%) once daily at night **for 7 days**	• Antibiotic treatment for BV is indicated for symptomatic women only • Routine screening for, or treatment of, asymptomatic women for BV to reduce perinatal complications is not indicated • Urine pH dipsticks are not suitable for diagnosis
Chlamydia trachomatis (CT) **Gonorrhoea: *Neisseria gonorrhoeae (NG)***	• Vaginal discharge • Dysuria without significant bacteriuria or persistent dysuria despite successful treatment of bacteriuria	**'High risk for STI'** • CT/NG screening in the first trimester: ST-LVS, urine for CT/NG PCR • ToC 3-4 weeks post-treatment • Re-testing in 3 months	**NG**: • IM ceftriaxone 1g **STAT** **CT**: PO azithromycin 1g **STAT**, followed by 500mg 24 hourly **for 2 days**	**NG:** • IM spectinomycin 2g **STAT** + PO azithromycin 2g **STAT**	• Please refer to GUM for partner management
Influenza Refer to management of influenza on Microguide Body system > Respiratory > Influenza	• Fever, cough, runny-nose, SOB, headache, myalgia	• Nasopharyngeal swab in VTM for Flu PCR	**First line:** PO oseltamivir 75mg twice daily **for 5 days** (for weight ≥ 41kg)	**Second line:** Zanamivir inhaler (Relenza Diskhaler®) 10mg 12 hourly for **5 days**; If confirmed oseltamivir resistant influenza or if there is evidence of gastrointestinal dysfunction (such as malabsorption, gastroparesis, uncontrollable vomiting and gastrointestinal bleeding).	Pregnant women with influenza virus infection are at greater risk of developing complicated and more severe influenza. PO oseltamivir and INH zanamivir are safe in pregnancy and with breast feeding
Malaria	• Fever, headache, malaise • GI disturbances: Nausea, abdominal pain, vomiting, diarrhoea **PLUS** • H/o travel to malaria-endemic area in	Thick and thin blood films Rapid diagnostic test 3 negative diagnostic samples over 24 hours are required to exclude malaria Blood film may be negative in complicated falciparum malaria due to high parasite load in placenta	**Uncomplicated malaria in the first trimester:** Quinine + Clindamycin PO **Uncomplicated malaria in the 2nd and 3rd trimester:** Artemether-lumefantrine (Riamet®) PO **Severe malaria in any trimester:** Artesunate IV		Refer to consultant medical microbiologist

Clinical Indication	Clinical Assessment	Diagnostic Evaluation	Treatment Recommendations		Comments
			First Choice	Alternatives e.g. 'True Penicillin allergy'	
	previous 6 days-6 months, regardless of antimalarial prophylaxis				

Appendix 1: IV Gentamicin Prescribing and Monitoring Algorithm in Obstetrics Patients

Exclusions

The algorithm does not apply to:

1. Stat doses of gentamicin – no monitoring is required
2. Post-partum patients – please refer to the Adult IV Gentamicin guideline
3. Child <16 years – refer to Paediatric Guidelines
4. Patients with the following contraindications to gentamicin use (please contact the Microbiology team for advice):
   - Myasthenia gravis
   - Acute Kidney Injury (AKI) stage 3
   - Chronic Kidney Disease (CKD) stage 5 not on dialysis.
   - Renal transplant
   - Hypersensitivity to gentamicin/aminoglycosides

Appendix 2: IV Gentamicin Prescribing and Monitoring Algorithm in Obstetrics Patients

RENAL IMPAIRMENT

Assess patient for underlying renal impairment and risk of AKI. Review most recent GFR and clinical risk factors.

GFR ≤ 30mL/min
OR
High risk of AKI

GFR > 30mL/min
AND
Low risk of AKI

CALCULATE ANTENATAL DOSE

Low-dose extended interval (LDEI) IV gentamicin regimen:

GFR (mL/min)	Regimen
20 – 30	2mg/kg 24hrly
15 – 20	2mg/kg 48hrly
Haemodialysis	2mg/kg 48hrly
< 15	Seek alternative

Check level between 20-24 hours of starting treatment

Prescribe dose:
5mg/kg
(pregnancy booking weight) – see table 5 below

If **BMI >30** at booking, use **corrected** body weight

Check level between 20-24 hours of starting treatment

If levels not back, give 2nd dose and chase levels.

Is the level <1mg/L?

NO

DO NOT give another dose. Re-check level after 12 hours.

YES

Continue treatment

MONITOR GENTAMICIN LEVELS AND RENAL FUNCTION

IS THE PATIENT POST-PARTUM?
Yes – refer to Adult Gentamicin Guideline

NO

Continue treatment.
Review by Senior Obstetric Staff daily with Microbiology advice.
Monitor renal function/ototoxicity daily.
If >7 days treatment required, consider referral to audiology.

Advice:
On Call Microbiologist: via Switchboard
Antimicrobial Pharmacist: Bleep 290 or Pager 40318
Clinical Pharmacist: bleep number on ward

Table 5: IV Gentamicin 5mg/kg dose banding table for patients with GFR > 30mL/min and low risk of AKI

Booking weight (kg)	Gentamicin dose
≤ 50 to 54	260mg
55 to 58	280mg
59 to 62	300mg
63 to 66	320mg
67 to 70	340mg
71 to 74	360mg
75 to 78	380mg
79 to 82	400mg
83 to 86	420mg
87 to 90	440mg
91 to 94	460mg
≥ 95	500mg

Doses rounded to the nearest 20mg to minimise vial wastage.
Maximum daily dose 500mg.

Table 6: BMI table for height in meter and weight in kilogram

Body Mass Index Table

	Normal						Overweight					Obese										Extreme obesity														
BMI	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Height (m)	**Body Weight (kg)**																																			
1.46	41	43	45	47	49	51	53	55	58	60	62	64	66	68	70	72	75	77	79	81	83	85	87	90	92	94	96	98	100	102	104	107	109	111	113	115
1.48	42	44	46	48	50	53	55	57	59	61	64	66	68	70	72	74	77	79	81	83	85	88	90	92	94	96	99	101	103	105	107	110	112	114	116	118
1.50	43	45	47	50	52	54	56	59	61	63	65	68	70	72	74	77	79	81	83	86	88	90	92	95	97	99	101	104	106	108	110	113	115	117	119	122
1.52	44	46	49	51	53	55	58	60	62	65	67	69	72	74	76	79	81	83	85	88	90	92	95	97	99	102	104	106	109	111	113	116	118	120	122	125
1.54	45	47	50	52	55	57	59	62	64	66	69	71	74	76	78	81	83	85	88	90	92	95	97	100	102	104	107	109	111	114	116	119	121	123	126	128
1.56	46	49	51	54	56	58	61	63	66	68	71	73	75	78	80	83	85	88	90	92	95	97	100	102	105	107	110	112	114	117	119	122	124	127	129	131
1.58	47	50	52	55	57	60	62	65	67	70	72	75	77	80	82	85	87	90	92	95	97	100	102	105	107	110	112	115	117	120	122	125	127	130	132	135
1.60	49	51	54	56	59	61	64	67	69	72	74	77	79	82	84	87	90	92	95	97	100	102	105	108	110	113	115	118	120	123	125	128	131	133	136	138
1.62	50	52	55	58	60	63	66	68	71	73	76	79	81	84	87	89	92	94	97	100	102	105	108	110	113	115	118	121	123	126	129	131	134	136	139	142
1.64	51	54	56	59	62	65	67	70	73	75	78	81	83	86	89	91	94	97	100	102	105	108	110	113	116	118	121	124	126	129	132	134	137	140	143	145
1.66	52	55	58	61	63	66	69	72	74	77	80	83	85	88	91	94	96	99	102	105	107	110	113	116	118	121	124	127	130	132	135	138	141	143	146	149
1.68	54	56	59	62	65	68	71	73	76	79	82	85	87	90	93	96	99	102	104	107	110	113	116	119	121	124	127	130	133	135	138	141	144	147	150	152
1.70	55	58	61	64	66	69	72	75	78	81	84	87	90	92	95	98	101	104	107	110	113	116	118	121	124	127	130	133	136	139	142	145	147	150	153	156
1.72	56	59	62	65	68	71	74	77	80	83	86	89	92	95	98	101	104	107	109	112	115	118	121	124	127	130	133	136	139	142	145	148	151	154	157	160
1.74	58	61	64	67	70	73	76	79	82	85	88	91	94	97	100	103	106	109	112	115	118	121	124	127	130	133	136	139	142	145	148	151	154	157	160	163
1.76	59	62	65	68	71	74	77	81	84	87	90	93	96	99	102	105	108	112	115	118	121	124	127	130	133	136	139	142	146	149	152	155	158	161	164	167
1.78	60	63	67	70	73	76	79	82	86	89	92	95	98	101	105	108	111	114	117	120	124	127	130	133	136	139	143	146	149	152	155	158	162	165	168	171
1.80	62	65	68	71	75	78	81	84	87	91	94	97	100	104	107	110	113	117	120	123	126	130	133	136	139	143	146	149	152	156	159	162	165	168	172	175
1.82	63	66	70	73	76	79	83	86	89	93	96	99	103	106	109	113	116	119	123	126	129	132	136	139	142	146	149	152	156	159	162	166	169	172	176	179
1.84	64	68	71	74	78	81	85	88	91	95	98	102	105	108	112	115	118	122	125	129	132	135	139	142	146	149	152	156	159	163	166	169	173	176	179	183
1.86	66	69	73	76	80	83	86	90	93	97	100	104	107	111	114	118	121	125	128	131	135	138	142	145	149	152	156	159	163	166	170	173	176	180	183	187
1.88	67	71	74	78	81	85	88	92	95	99	102	106	110	113	117	120	124	127	131	134	138	141	145	148	152	156	159	163	166	170	173	177	180	184	187	191
1.90	69	72	76	79	83	87	90	94	97	101	105	108	112	116	119	123	126	130	134	137	141	144	148	152	155	159	162	166	170	173	177	181	184	188	191	195
1.92	70	74	77	81	85	88	92	96	100	103	107	111	114	118	122	125	129	133	136	140	144	147	151	155	159	162	166	170	173	177	181	184	188	192	195	199
1.94	72	75	79	83	87	90	94	98	102	105	109	113	117	120	124	128	132	135	139	143	147	151	154	158	162	166	169	173	177	181	184	188	192	196	199	203
1.96	73	77	81	85	88	92	96	100	104	108	111	115	119	123	127	131	134	138	142	146	150	154	158	161	165	169	173	177	181	184	188	192	196	200	204	207
1.98	74	78	82	86	90	94	98	102	106	110	114	118	122	125	129	133	137	141	145	149	153	157	161	165	169	172	176	180	184	188	192	196	200	204	208	212
2.00	76	80	84	88	92	96	100	104	108	112	116	120	124	128	132	136	140	144	148	152	156	160	164	168	172	176	180	184	188	192	196	200	204	208	212	216

Table 7: BMI table for heights in inches and weight in pounds

Body Mass Index Table

	Normal						Overweight					Obese										Extreme Obesity														
BMI	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Height (inches)																Body Weight (pounds)																				
58	91	96	100	105	110	115	119	124	129	134	138	143	148	153	158	162	167	172	177	181	186	191	196	201	205	210	215	220	224	229	234	239	244	248	253	258
59	94	99	104	109	114	119	124	128	133	138	143	148	153	158	163	168	173	178	183	188	193	198	203	208	212	217	222	227	232	237	242	247	252	257	262	267
60	97	102	107	112	118	123	128	133	138	143	148	153	158	163	168	174	179	184	189	194	199	204	209	215	220	225	230	235	240	245	250	255	261	266	271	276
61	100	106	111	116	122	127	132	137	143	148	153	158	164	169	174	180	185	190	195	201	206	211	217	222	227	232	238	243	248	254	259	264	269	275	280	285
62	104	109	115	120	126	131	136	142	147	153	158	164	169	175	180	186	191	196	202	207	213	218	224	229	235	240	246	251	256	262	267	273	278	284	289	295
63	107	113	118	124	130	135	141	146	152	158	163	169	175	180	186	191	197	203	208	214	220	225	231	237	242	248	254	259	265	270	278	282	287	293	299	304
64	110	116	122	128	134	140	145	151	157	163	169	174	180	186	192	197	204	209	215	221	227	232	238	244	250	256	262	267	273	279	285	291	296	302	308	314
65	114	120	126	132	138	144	150	156	162	168	174	180	186	192	198	204	210	216	222	228	234	240	246	252	258	264	270	276	282	288	294	300	306	312	318	324
66	118	124	130	136	142	148	155	161	167	173	179	186	192	198	204	210	216	223	229	235	241	247	253	260	266	272	278	284	291	297	303	309	315	322	328	334
67	121	127	134	140	146	153	159	166	172	178	185	191	198	204	211	217	223	230	236	242	249	255	261	268	274	280	287	293	299	306	312	319	325	331	338	344
68	125	131	138	144	151	158	164	171	177	184	190	197	203	210	216	223	230	236	243	249	256	262	269	276	282	289	295	302	308	315	322	328	335	341	348	354
69	128	135	142	149	155	162	169	176	182	189	196	203	209	216	223	230	236	243	250	257	263	270	277	284	291	297	304	311	318	324	331	338	345	351	358	365
70	132	139	146	153	160	167	174	181	188	195	202	209	216	222	229	236	243	250	257	264	271	278	285	292	299	306	313	320	327	334	341	348	355	362	369	376
71	136	143	150	157	165	172	179	186	193	200	208	215	222	229	236	243	250	257	265	272	279	286	293	301	308	315	322	329	338	343	351	358	365	372	379	386
72	140	147	154	162	169	177	184	191	199	206	213	221	228	235	242	250	258	265	272	279	287	294	302	309	316	324	331	338	346	353	361	368	375	383	390	397
73	144	151	159	166	174	182	189	197	204	212	219	227	235	242	250	257	265	272	280	288	295	302	310	318	325	333	340	348	355	363	371	378	386	393	401	408
74	148	155	163	171	179	186	194	202	210	218	225	233	241	249	256	264	272	280	287	295	303	311	319	326	334	342	350	358	365	373	381	389	396	404	412	420
75	152	160	168	176	184	192	200	208	216	224	232	240	248	256	264	272	279	287	295	303	311	319	327	335	343	351	359	367	375	383	391	399	407	415	423	431
76	156	164	172	180	189	197	205	213	221	230	238	246	254	263	271	279	287	295	304	312	320	328	336	344	353	361	369	377	385	394	402	410	418	426	435	443

Source: Adapted from *Clinical Guidelines on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults: The Evidence Report.*

Table 8: Dosing weight table

Plot booking weight and height in the table to determine the Dosing Weight (in kg). Then use this dosing weight to calculate gentamicin dose. If the value falls between two points on the table, take the mid-point (average) dosing weight. *Note: the table displays a corrected dosing weight when BMI ≥ 30kg/m^2.*

Booking weight (stone & pounds)

Patient height (cm)	5st 7	6st 4	7st 1	7st 12	8st 9	9st 6	10st 3	11st 0	11st 11	12st 8	13st 5	14st 2	14st 13	15st 10	16st 7	17st 5	18st 2	18st 13	19st 10	20st 7	21st 4	22st 1	Patient height (feet & inches)
191	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	91	93	95	97	99	101	103	**6' 3''**
185	35	40	45	50	55	60	65	70	75	80	85	90	95	100	86	88	90	92	94	96	98	100	**6' 1''**
183	35	40	45	50	55	60	65	70	75	80	85	90	95	100	85	87	89	91	93	95	97	99	**6' 0''**
180	35	40	45	50	55	60	65	70	75	80	85	90	95	81	83	85	87	89	91	93	95	97	**5' 11''**
175	35	40	45	50	55	60	65	70	75	80	85	90	77	79	81	83	85	87	89	91	93	95	**5' 9''**
173	35	40	45	50	55	60	65	70	75	80	85	74	76	78	80	82	84	86	88	90	92	94	**5' 8''**
170	35	40	45	50	55	60	65	70	75	80	85	72	74	76	78	80	82	84	86	88	90	92	**5' 7''**
165	35	40	45	50	55	60	65	70	75	80	67	69	71	73	75	77	79	81	83	85	87	89	**5' 5''**
163	35	40	45	50	55	60	65	70	75	64	66	68	70	72	74	76	78	80	82	84	86	88	**5' 4''**
157	35	40	45	50	55	60	65	70	59	61	63	65	67	69	71	73	75	77	79	81	83	85	**5' 2''**
155	35	40	45	50	55	60	65	70	58	60	62	64	66	68	70	72	74	76	78	80	82	84	**5' 1''**
152	35	40	45	50	55	60	65	59	61	63	65	67	69	71	73	75	77	79	81	83	85	87	**5' 0''**
147	35	40	45	50	55	60	55	57	59	61	63	65	67	69	71	73	75	77	79	81	83	85	**4' 10''**
145	35	40	45	50	55	60	54	56	58	60	62	64	66	68	70	72	74	76	78	80	82	84	**4' 9''**
142	35	40	45	50	55	60	53	55	57	59	61	63	65	67	69	71	73	75	77	79	81	83	**4' 8''**
137	35	40	45	50	55	49	51	53	55	57	59	61	63	65	67	69	71	73	75	77	79	81	**4' 6''**
135	35	40	45	50	47	49	51	53	55	57	59	61	63	65	67	69	71	73	75	77	79	81	**4' 5''**
130	35	40	45	50	45	47	49	51	53	55	57	59	61	63	65	67	69	71	73	75	77	79	**4' 3''**
127	35	40	45	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70	72	74	76	78	**4' 2''**
	35	**40**	**45**	**50**	**55**	**60**	**65**	**70**	**75**	**80**	**85**	**90**	**95**	**100**	**105**	**110**	**115**	**120**	**125**	**130**	**135**	**140**	

Booking weight (kg)

Appendix 3: IV Teicoplanin 10mg/kg dose banding table

Booking weight	Teicoplanin dose
40kg or less	400mg
41kg – 58kg	600mg
59kg – 74kg	800mg
75kg – 91kg	1000mg
92kg – 108kg	1200mg
109kg – 124kg	1400mg
125kg – 141kg	1600mg
142kg – 158kg	1800mg
159kg – 174kg	2000mg
>175kg	12mg/kg Dose rounded to the nearest 200mg vial

Doses rounded to the nearest 200mg to minimise vial wastage.

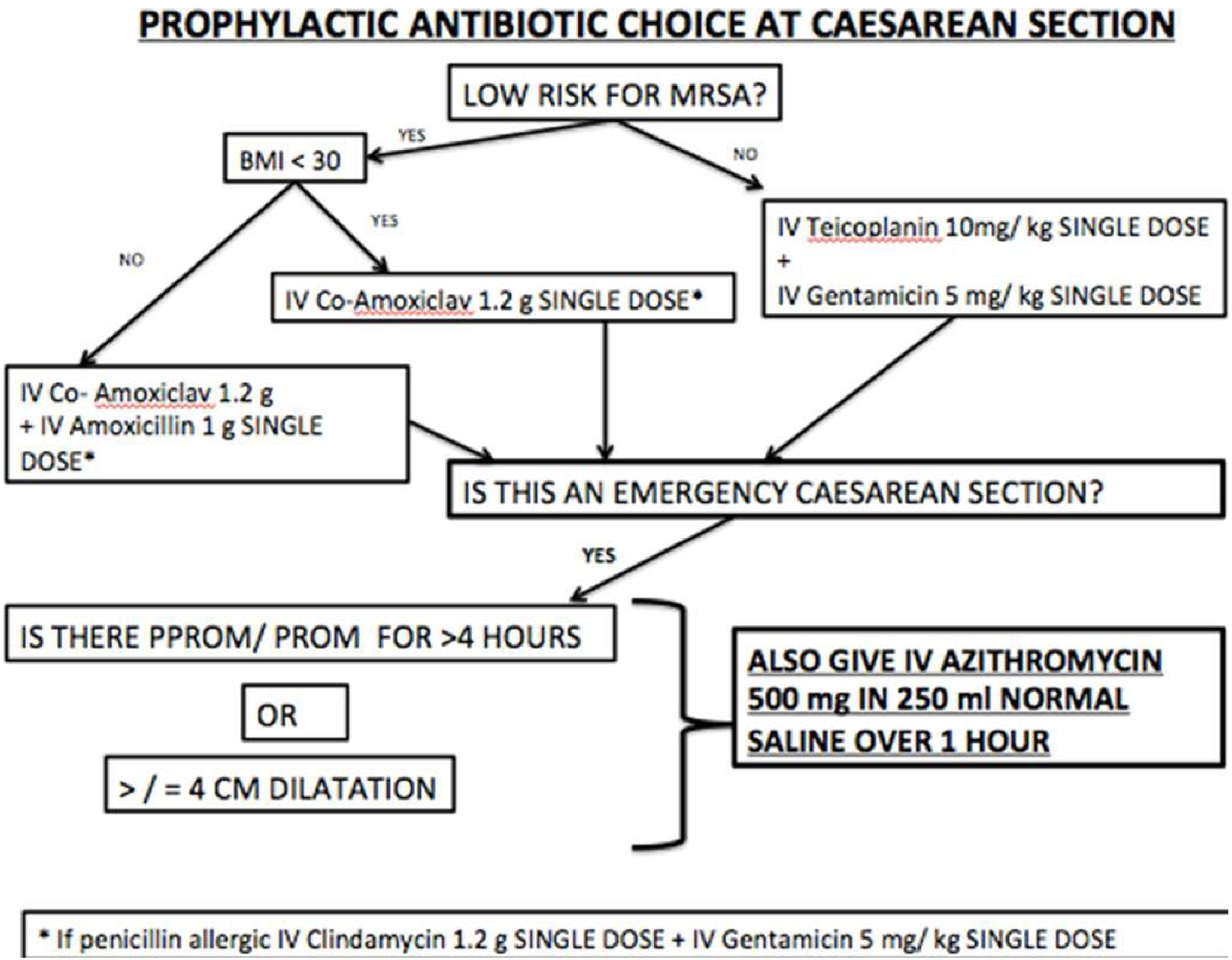
For patients with renal impairment, please see the table below:

CrCl (mL/min)	Loading dose	Maintenance dose
Between 30 – 80mL/min	Normal dose regimen on days 1 – 4. **See table above**	Normal maintenance dose every 48hrs. **See table above**
Less than 30mL/min and in haemodialysis		Normal maintenance dose every 72hrs. **See table above**

Appendix 4: Antibiotic costs for obstetric antibiotic guideline

Antibiotic	Cost per unit (£)	Average cost per day per patient
Amoxicillin IV 1g 8 hourly	0.77 / 1000mg	£2.31
Amoxicillin PO 1g 8 hourly	0.10 / 500mg	£0.62
Azithromycin PO 1g STAT	0.26 / 250mg	£1.04
Benzylpenicillin IV 3g STAT	5.26 / 1.2g	£15.78
Benzylpenicillin IV 1.5g 4 hourly	5.26 / 1.2g	£63.12
Benzylpenicillin IV 1.2g 4 hourly	5.26 / 1.2g	£31.56
Cefradine PO 500mg 6 hourly	0.42 / 500mg	£1.68
Ceftriaxone IV 2g 24 hourly	1.20 / 2g	£1.20
Ceftriaxone IM 500mg STAT	0.48 / 1g	£0.48
Clarithromycin IV 500mg 12 hourly	1.64 / 500mg	£3.28
Clarithromycin PO 500mg 12 hourly	0.17 / 500mg	£0.34
Clindamycin IV 1.2g 6 hourly	1.08 / 600mg	£8.64
Clindamycin PO 450mg 6 hourly	0.07 / 150mg	£0.84
Clindamycin intravaginal cream (2%)	13.03 / 40g	£13.03
Co-amoxiclav IV 1.2g 8 hourly	0.91 / 1.2g	£2.73
Co-amoxiclav PO 625mg 8 hourly	0.09 / 625mg	£0.27
Fosfomycin PO 3g STAT	3.90 / 3g	£3.90
Gentamicin IV 5mg/kg 24 hourly (based on average 70kg)	0.41 / 80mg	£2.05
Metronidazole IV 500mg 8 hourly	0.55 / 500mg	£1.65
Metronidazole PO 400mg 8 hourly	0.03 / 400mg	£0.09
Metronidazole 500mg suppository STAT	1.82 / 500mg	£1.82
Metronidazole intravaginal gel (0.75%)	10.43 / 40g	£10.43
Nitrofurantoin PO 100mg 6 hourly	0.45 / 50mg	£3.60
Piperacillin/tazobactam IV 4.5g 8 hourly	1.72 / 4.5g	£5.16
Spectinomycin IM 2g STAT	33.53 / 2g	£33.53
Teicoplanin IV 10mg/kg (12 hourly for 3 doses then 24 hourly (based on average 70kg)	6.84 / 400mg	£6.00
Trimethoprim PO 100mg 12 hourly	0.01 / 100mg	£0.02

Appendix 5: Prophylactic antibiotic choice at Caesarean section



Appendix 6: Risk assessment for women at risk of preterm birth

Intermediate risk	High risk
Previous birth by caesarean section at full dilatation	Previous preterm birth of mid-trimester loss (16 to 34 weeks gestation)
	Previous preterm prelabour rupture of membranes <34/40
	Previous use of cervical cerclage
History of significant cervical excisional event i.e., large loop excision of the transformation zone (LLETZ) where >15mm depth removed, or >1 LLETZ procedure carried out or cone biopsy (knife or laser, typically carried out under general anaesthetic)	Known uterine variant (i.e., unicornuate, bicornuate uterus or uterine septum)
	Intrauterine adhesions (Ashermann's syndrome)
	History of trachelectomy (for cervical cancer)

The assessment should take place at the booking appointment with referral by 12 weeks.

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